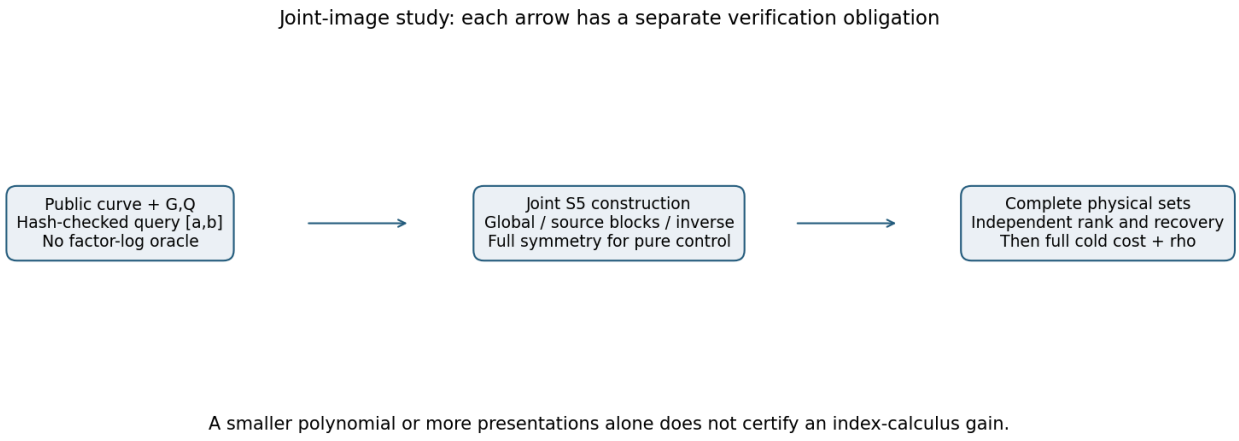
Repaired finite-count proof passes

Gate uniform-lean-v3 checks the repaired source in ResidualMass.lean (../../proof/uniform-v1/ResidualMass.lean). The exact launch and finish timestamps, source hash, source snapshot and command are retained in runs/uniform-lean-v3/metadata/.



The build and transitive-axiom gate pass. The audit covers 12 declarations: the six named finite-count theorems and generated recursive equations. There are zero failing and zero trusted-native declarations. Standard foundations in the public theorem proofs are propext and Quot.sound. The loose audit log (../../results/proof-checks/uniform-v3/audit.log) and build/status files preserve this result.

The theorem proves that all finite buckets together contain exactly the candidate-list length when labels are bounded by n. Applying two bounded labellings to the same list therefore gives equal total mass. It does not assume or prove independent sampling, rank, a physical domain-size formula, curve-map disjointness or runtime. Those are separate bridges in the proof map (../../proof/PROOF_MAP.md).

The command uses pinned Lean 4.34.1, one worker, 8 GiB, a 900 s watchdog and the busy reservation. This is correctness evidence; no timing comparison is admitted and the visual is explanatory.